

RENMARK AP, Australia

WMO: 956870

Lat: 34.1983S

Lon: 140.6767E

Elev: 105

StdP: 14.64

Time Zone: 9.50 (AUA)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.0	33.0	18.7	14.1	71.8	21.9	16.5	65.8	25.3	60.6	22.8	60.5	1.5	260	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.0	104.7	65.8	100.3	64.7	96.3	63.7	71.0	85.6	68.9	86.9	67.2	86.8	12.3	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.3	74.2	64.1	89.8	73.1	60.9	80.2	72.1	34.9	84.5	33.1	86.7	31.7	86.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.6	18.3	DB	24.8	111.3	1.6	2.5	23.6	113.1	22.6	114.5	21.7	115.9	20.5	117.7	(4)
(5)			WB	23.9	73.6	1.7	2.6	22.7	75.5	21.7	77.0	20.8	78.4	19.5	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	76.6	75.4	69.7	62.3	55.6	50.7	50.0	52.0	57.1	62.5	68.4	72.3		
	DBStd	11.28	7.64	7.57	7.18	6.26	5.26	4.73	4.60	4.97	6.10	6.91	7.79	7.49		
	HDD50	148	0	0	0	1	9	47	57	31	3	0	0	0		
	HDD65	2198	2	5	29	124	294	429	464	403	252	136	48	12		
	CDD50	4764	825	711	610	369	184	69	58	93	217	386	551	691		
	CDD65	1340	363	296	174	43	4	0	0	1	15	57	148	239		
	CDH74	19021	5050	3867	2269	659	59	1	2	35	305	1059	2226	3489		
	CDH80	10062	2957	2154	1108	232	5	0	0	4	96	467	1139	1900		
(14) Wind	WSAvg	8.9	10.2	10.1	9.0	7.2	7.3	7.2	8.2	8.6	9.3	9.8	10.0	10.2		
(15) Precipitation	PrecAvg	10.2	0.7	0.7	0.5	0.7	0.9	0.8	1.1	1.0	1.1	1.2	0.9	0.7		
	PrecMax	20.4	3.8	4.1	3.2	5.2	3.6	2.4	2.8	3.0	4.4	4.3	4.3	2.7		
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0		
	PrecStd	3.5	0.7	0.9	0.6	0.9	0.7	0.6	0.6	0.6	0.9	1.1	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	110.0	108.0	101.4	93.2	80.1	70.7	70.8	78.3	88.3	97.0	103.9	107.4		
		MCWB	67.7	67.9	65.1	62.1	57.5	55.5	53.0	54.6	57.8	60.3	64.0	64.9		
	2%	DB	105.4	102.0	96.2	86.4	74.8	65.8	66.1	72.4	82.5	90.8	97.1	101.4		
		MCWB	66.5	66.2	63.8	60.4	57.0	53.4	52.1	52.7	56.3	59.5	62.4	63.9		
	5%	DB	100.2	97.2	91.5	81.4	70.5	62.9	62.8	67.5	76.5	85.3	91.8	96.2		
		MCWB	65.1	65.1	63.6	58.7	56.0	52.8	51.5	51.5	55.2	57.9	61.6	62.9		
	10%	DB	94.9	92.6	86.2	76.8	67.0	60.6	60.1	63.8	71.5	79.6	86.5	90.7		
		MCWB	64.5	64.2	62.1	57.8	54.9	51.9	50.6	50.7	53.9	56.7	60.0	61.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	74.0	69.2	66.3	61.6	58.7	57.0	57.3	61.2	64.8	68.9	71.0		
		MCDB	82.0	86.7	84.1	70.2	70.4	63.1	62.2	69.5	72.6	81.3	83.2	82.4		
	2%	WB	70.3	72.1	67.3	63.1	59.3	55.8	54.3	54.8	58.6	62.0	66.3	68.0		
		MCDB	88.5	85.0	84.5	77.3	68.5	62.2	61.7	66.7	74.6	78.8	84.3	82.7		
	5%	WB	68.5	69.5	65.5	61.3	57.5	54.2	52.7	53.3	56.8	60.2	64.6	65.8		
		MCDB	90.4	86.5	84.1	76.5	67.2	60.8	60.2	63.8	73.3	77.5	83.6	87.0		
	10%	WB	66.6	66.9	63.8	59.5	56.0	52.7	51.4	51.8	55.1	58.4	62.5	63.8		
		MCDB	90.1	87.2	83.1	74.0	65.1	59.2	58.6	62.0	68.9	76.0	80.8	86.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.0	27.4	26.2	26.0	22.3	20.5	20.5	23.1	25.6	27.4	27.8	28.6		
		MCDBR	36.8	34.3	33.7	33.9	27.7	23.3	24.0	30.5	34.4	37.3	36.0	36.9		
		MCWBR	12.1	11.0	12.1	14.8	13.9	13.5	13.1	15.2	14.9	14.2	12.2	12.5		
	5% WB	MCDBR	27.0	24.0	26.7	24.7	21.9	19.2	20.1	24.1	28.1	27.9	27.7	26.9		
		MCWBR	10.4	9.2	10.5	11.7	11.9	11.7	11.9	13.0	13.0	11.6	10.3	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.338	0.320	0.312	0.299	0.292	0.291	0.302	0.333	0.333	0.343	0.334		
	taud		2.498	2.532	2.566	2.563	2.583	2.591	2.589	2.543	2.445	2.463	2.468	2.507		
	Ebn at Noon		316	310	305	289	277	270	277	289	295	308	313	319		
	Edh at Noon		36	34	30	27	23	22	23	27	34	36	37	36		
(44) All-Sky Solar Radiation	RadAvg		2511	2247	1858	1361	958	784	846	1150	1561	1997	2285	2495		
	RadStd		121	122	98	69	45	46	37	75	110	122	107	93		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	-22	-166	+272	+117

Nomenclature: See separate page